ch21 Circuit and Power

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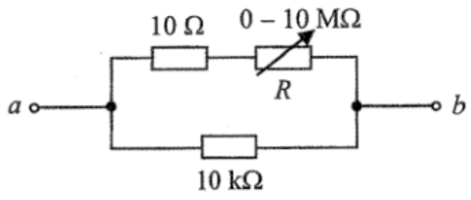
2016 Q25

25.



In the above circuit, the variable resistor R can be adjusted over its full range from 0 to 10 MΩ. What is the approximate range of resistance between a and b ?

A. 0 to 10 kΩ
B. 10 Ω to 10 kΩ
C. 10 Ω to 10 MΩ
D. 10 kΩ to 10 MΩ

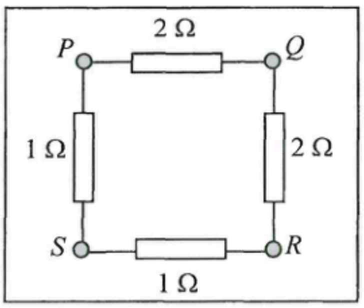
2018 Q26

26. A mobile phone battery labelled 2800 mA h capacity is fully charged initially. What percentage of capacity remains after it has delivered a current of 200 mA for 3 hours ?

A. 7.1%
B. 21.4%
C. 78.6%
D. 92.9%

2024 Q24

24.



Four resistors are connected to terminals $PQRS$ of a circuit board as shown. Across which pair of terminals below is the equivalent resistance smallest ?

A. PQ
B. PR
C. PS
D. QS

2020 Q25

25.

The battery shown has a capacity of 1100 mA h. How much energy is delivered when the battery operates normally at a current of 250 mA for one hour ? Assume that the battery's operating voltage remains at 3.7 V during that period.

A. $(3.7 \times \frac{250}{1000} \times 3600)$ J

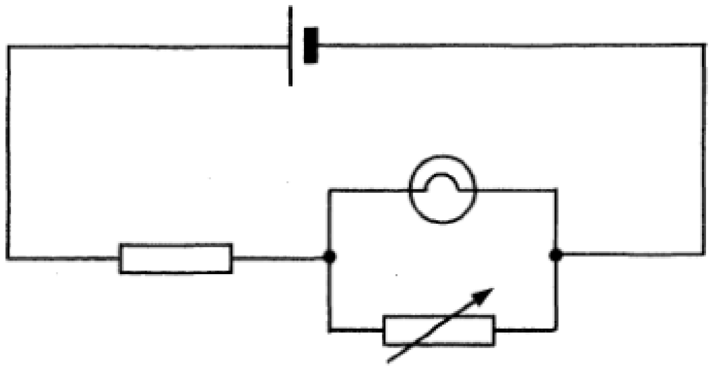
B. $(3.7 \times \frac{1100}{1000} \times 3600)$ J

C. $(3.7 \times \frac{250}{1000} \times 1)$ J

D. $(3.7 \times \frac{1100}{1000} \times 1)$ J

2012 Q27

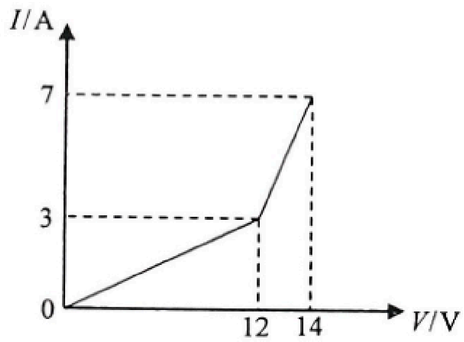
27. What will happen if the variable resistor is set to zero in the circuit below ?



A. The light bulb will burn out.
B. The light bulb will not light up.
C. The brightness of the light bulb will increase.
D. The brightness of the light bulb will remain unchanged.

2018 Q23

23. The graph below shows the current-voltage (I-V) relationship of a conductor.

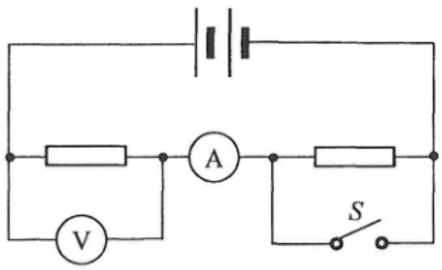


Which statement below is **INCORRECT** ?

A. When the voltage across the conductor is less than 12 V, it obeys Ohm's law.
B. When the voltage across the conductor exceeds 12 V, its resistance begins to decrease.
C. The resistance of the conductor is 0.5 Ω when the current flowing through it is 5 A.
D. The resistance of the conductor is 2 Ω when the voltage across it is 14 V.

2024 Q25

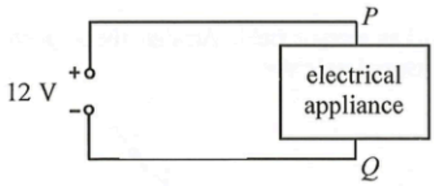
25. In the circuit below, the two resistors are identical. When S is closed, how would the readings of the ammeter and voltmeter change ?



	ammeter reading	**voltmeter reading**
A.	increase	increase
B.	increase	decrease
C.	decrease	increase
D.	decrease	decrease

2023 Q26

26. The figure shows an electrical appliance connected to a potential difference of 12 V.



When a charge of +2 C passes from P to Q in the circuit, how much electrical energy is supplied to the appliance ?

A. 2 J
B. 6 J
C. 12 J
D. 24 J

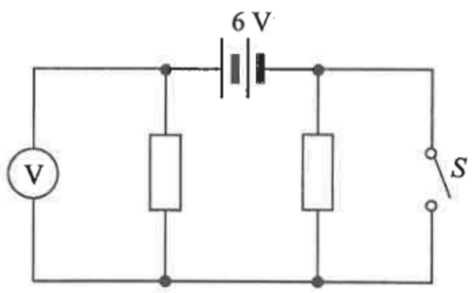
2015 Q27

*27. A copper wire of uniform cross-section carries a current of 0.5 A. Each metre of the wire contains 10^{22} free electrons. Find the magnitude of the drift velocity of the electrons in the wire.

A. 2.5×10^{-3} m s^{-1}
B. 7.8×10^{-3} m s^{-1}
C. 3.1×10^{-4} m s^{-1}
D. 9.7×10^{-4} m s^{-1}

2022 Q26

26.

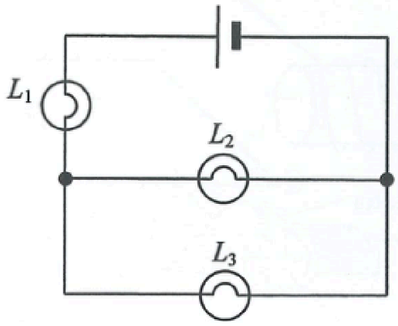


In the above circuit, the resistors are identical and the 6 V battery has negligible internal resistance. Which of the following is the set of readings on the voltmeter when (1) S is open; (2) S is closed ?

	***S* open**	***S* closed**
A.	0 V	6 V
B.	3 V	6 V
C.	6 V	0 V
D.	6 V	3 V

2019 Q25

25.



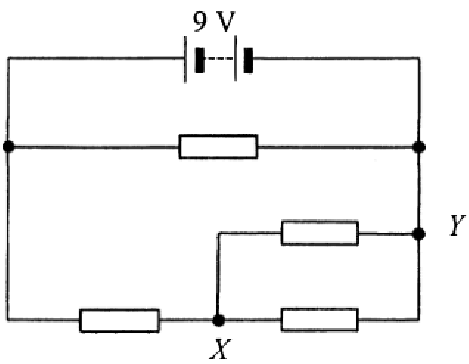
In the above circuit, L_1, L_2 and L_3 are three light bulbs and the cell has negligible internal resistance. Which of the following changes will make L_3 brighter ?

(1) L_1 is faulty and becomes a short circuit.
(2) L_2 is faulty and becomes a short circuit.
(3) L_2 is faulty and becomes an open circuit.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2017 Q24

24. In the circuit, all resistors are identical. The internal resistance of the battery can be neglected.

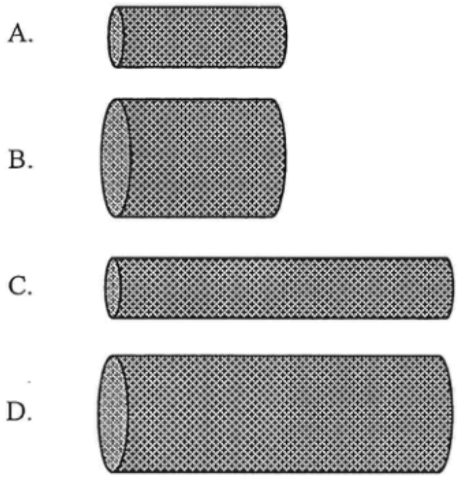


What is the potential difference between X and Y ?

A. 1.5 V
B. 3.0 V
C. 4.5 V
D. 6.0 V

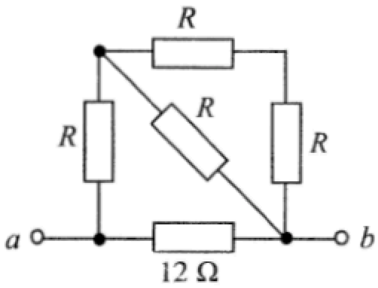
2020 Q22

22. The cylindrical resistors below are made from the same metal. Which one would produce the most power when the same voltage is applied in turns across the two ends of each resistor ?



2012 Q26

26.



In the above network, the resistance across terminals a and b is 6 Ω. If the 12 Ω resistor is replaced by a 6 Ω resistor, the resistance across terminals a and b

A. becomes 2 Ω.
B. becomes 4 Ω.
C. becomes 6 Ω.
D. cannot be found as the value of R is unknown.

2025 Q25

25. A student tests an electric circuit by connecting a filament light bulb to a dry cell using two connecting wires. The bulb does not light up. Which of the following is certainly **NOT** a possible explanation ?

A. The dry cell may be flat.
B. The circuit may be open due to a bad connection.
C. The dry cell may have its polarities connected the wrong way round.
D. The bulb may have blown.

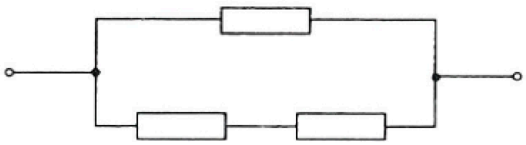
2022 Q25

25. An electrical device is charged with a steady d.c. of 60 mA for 30 minutes. Find the **number of electrons** that pass through the device during charging.

A. 108
B. 1800
C. 3.75×10^{17}
D. 6.75×10^{20}

2018 Q24

24.



Three identical resistors are arranged as shown. The rated power of each resistor is 12 W. If no resistor exceeds its rated power, what is the maximum power dissipation in such an arrangement ?

A. 16 W
B. 18 W
C. 20 W
D. 24 W

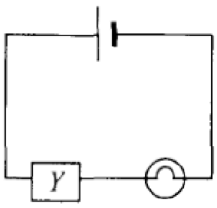
2018 Q25

25. Two wires X and Y of the same length are made from different materials. The radius of X is half that of Y. When the two wires are connected in parallel to the same power supply, the current flowing through each wire is the same. What is the ratio of the resistivity of the material of X to that of Y ?

A. 1 : 4
B. 4 : 1
C. 1 : 2
D. 2 : 1

2021 Q27

27. A light bulb is connected in series with a device Y and a cell as shown. Assume that the internal resistance of the cell is negligible and its e.m.f. remains unchanged.



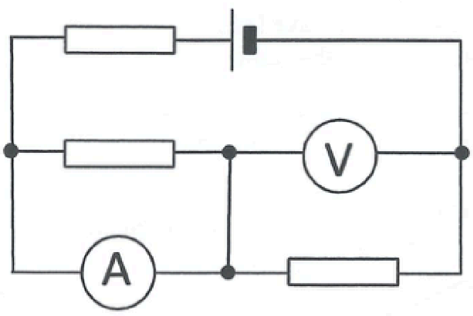
It is found that the brightness of the light bulb decreases with time. Which deductions must be correct ?

(1) The current in Y decreases with time.
(2) The voltage across Y decreases with time.
(3) The power supplied by the cell decreases with time.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2019 Q24

24.

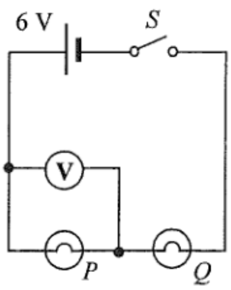


The figure shows a 6 V cell of negligible internal resistance connected to three identical resistors. Both ammeter and voltmeter are ideal. Find the voltmeter reading.

A. 6 V
B. 4 V
C. 3 V
D. 2 V

2014 Q25

25.

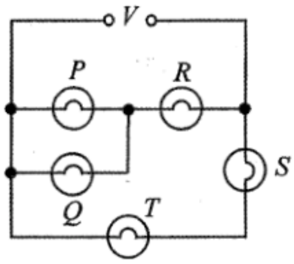


The figure shows two light bulbs P and Q connected to a cell of e.m.f. 6 V and negligible internal resistance. The voltmeter reads 6 V when the switch S is closed. Which of the following is possible ?

A. Both P and Q are short-circuited.
B. Both P and Q are burnt out and become open circuit.
C. P is short-circuited or Q is burnt out and becomes open circuit.
D. P is burnt out and becomes open circuit or Q is short-circuited.

2016 Q27

27.

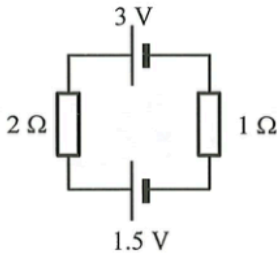


In the above circuit, all the bulbs are identical. If the voltage V gradually increases, which bulb(s) will burn out first ?

A. P and Q
B. R
C. S
D. T

2023 Q25

25. Two cells of negligible internal resistance are connected to two resistors as shown.

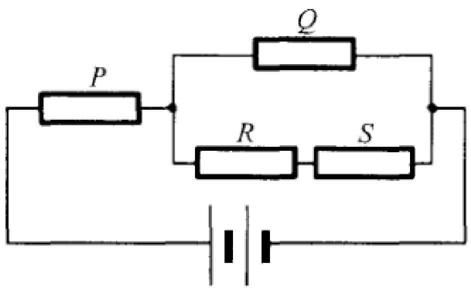


Which of the following is correct ?

	direction of current	current in the circuit
A.	anticlockwise	0.5 A
B.	clockwise	0.5 A
C.	anticlockwise	1.5 A
D.	clockwise	1.5 A

2021 Q23

23. Four identical resistors P, Q, R and S are connected to a battery of negligible internal resistance as shown.

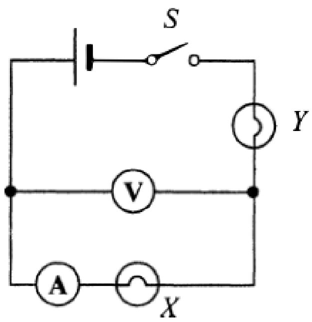


If the power dissipated by R is 1 W, find the total power output of the battery.

A. 11 W
B. 15 W
C. 19 W
D. 21 W

2013 Q32

32.



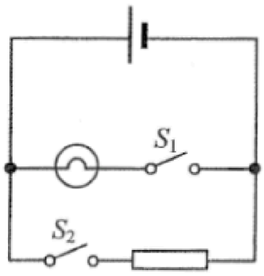
In the above circuit, the cell has negligible internal resistance. When switch S is closed, both bulbs are not lit. The voltmeter has a reading but the ammeter reads zero. If only one fault has been developed in the circuit, which of the following is possible ?

A. Bulb X has been shorted accidentally.
B. Bulb Y has been shorted accidentally.
C. Bulb X is burnt out and becomes open circuit.
D. Bulb Y is burnt out and becomes open circuit.

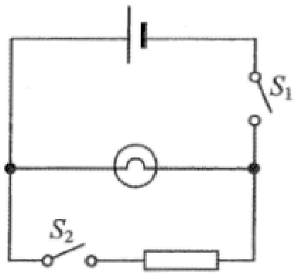
2015 Q25

25. For safety purposes, the driver seat of a car is equipped with a seat belt warning light. When the driver seat is occupied, the switch S_1 under his seat will close. If the seat belt is not yet fastened, switch S_2 will remain open and the warning light will light up. If the seat belt is fastened, the switch S_2 will close and the warning light will shut off. Which circuit design below is the best design ?

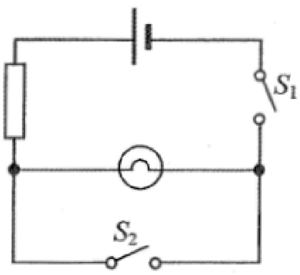
A.



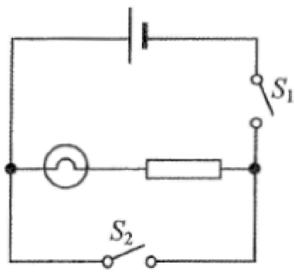
B.



C.

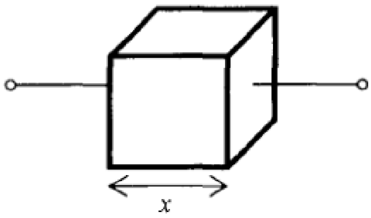


D.



2021 Q24

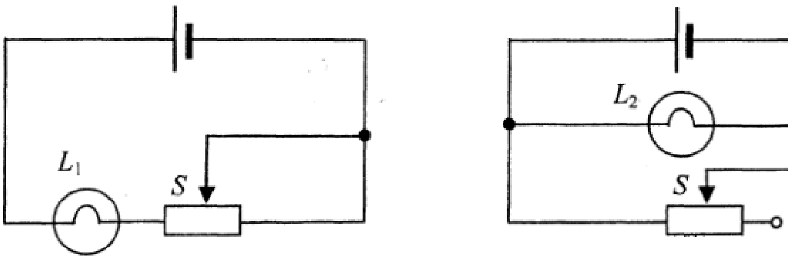
24. The figure shows a metallic cube of side length x. How is its resistance R between any two opposite faces related to x ?



A. $R \propto \frac{1}{x}$
B. $R \propto x$
C. $R \propto x^2$
D. $R \propto \frac{1}{x^2}$

2013 Q31

31.

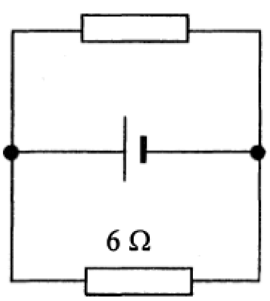


In each of the above circuits, the cell has constant e.m.f. and negligible internal resistance. When the sliding contact S of each rheostat shifts from the mid-position to the right, how would the brightness of each bulb change ?

	bulb L_1	bulb L_2
A.	becomes dimmer	remains unchanged
B.	becomes dimmer	becomes brighter
C.	remains unchanged	becomes dimmer
D.	becomes brighter	remains unchanged

2012 Q28

28.

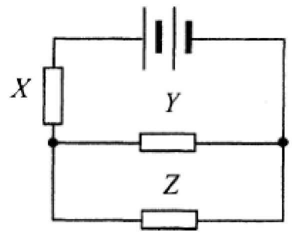


In the above circuit, the cell has e.m.f. 12 V and internal resistance 2 Ω. What is the current in the 6 Ω resistor ?

A. 0.5 A
B. 1.0 A
C. 1.5 A
D. 2.0 A

2013 Q30

30.



Resistors X, Y and Z in the above circuit are identical while the battery of negligible internal resistance supplies a total power of 24 W. What is the power dissipated in resistor Z ?

A. 3 W
B. 4 W
C. 6 W
D. 8 W

2016 Q26

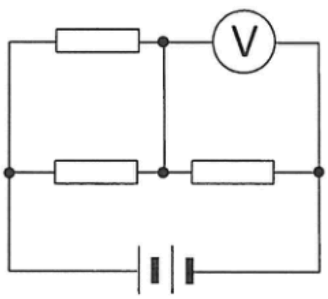
26. Two filament light bulbs X and Y are connected in parallel to a dry cell. X is brighter than Y. Which statements are correct ?

(1) In 1 s, the number of charges flowing through X is greater than that flowing through Y.
(2) In 1 s, the electrical energy dissipated by X is greater than that dissipated by Y.
(3) For every unit charge passing, the electrical energy dissipated by X is equal to that dissipated by Y.

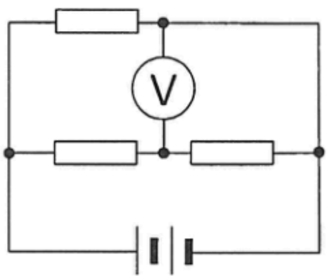
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2020 Q23

23. Three identical resistors, a battery of negligible internal resistance and an ideal voltmeter are connected to form Circuits (a) and (b) respectively.



Circuit (a)



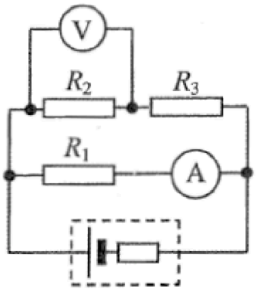
Circuit (b)

Given that the voltmeter reading is 8 V in Circuit (a), what is the voltmeter reading in Circuit (b) ?

A. 4 V
B. 6 V
C. 8 V
D. 12 V

2015 Q26

26. In the following circuit, the cell has a finite internal resistance and both meters are ideal.

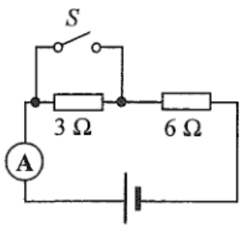


In which situation below will the readings of the ammeter and the voltmeter suddenly increase ?

A. R_1 is faulty and becomes a short circuit.
B. R_2 is faulty and becomes a short circuit.
C. R_3 is faulty and becomes a short circuit.
D. R_2 is faulty and becomes an open circuit.

2014 Q24

24.

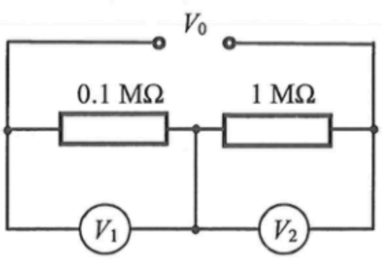


In the above circuit, the cell has constant e.m.f. and a fixed internal resistance. When S is closed, the ammeter reads 3.0 A. When S is open, which of the following is a possible reading of the ammeter ?

A. 1.6 A
B. 2.0 A
C. 2.4 A
D. 3.2 A

2025 Q26

26. In the circuit below, a voltage V_0 is applied across resistors of 0.1 MΩ and 1 MΩ connected in series. Voltmeter V_1 and V_2 are connected respectively across the two resistors as shown. V_1 is ideal while V_2 has an internal resistance of 1 MΩ.

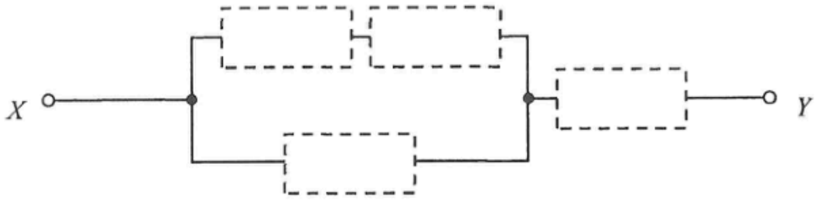


Estimate the ratio of the voltmeter readings $V_1 : V_2$.

A. 5 : 1
B. 1 : 5
C. 10 : 1
D. 1 : 10

2026 Q22

22.

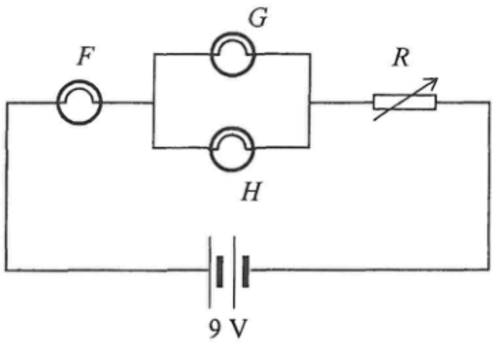


Four resistors of resistance 10 Ω, 20 Ω, 30 Ω and 40 Ω respectively are installed in the empty places in the above circuit. What is the minimum possible equivalent resistance across X and Y ?

A. 19.2 Ω
B. 25.6 Ω
C. 28.8 Ω
D. 32.2 Ω

2026 Q23

23.

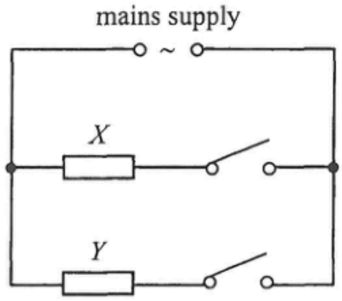


In the circuit above, light bulbs F, G and H are identical and the battery is ideal. Rheostat R is initially set to maximum resistance. The maximum voltage across each bulb without burning out is 2 V, and a burnt-out bulb becomes an open circuit. Initially all bulbs light up. If the resistance of R is gradually reduced to zero, which of the bulbs will burn out in the process ?

A. only bulb F
B. only bulbs G and H
C. none of them
D. all of them

2026 Q24

24.



The above shows a simplified diagram of the circuit in a heater. It consists of two heating elements X and Y. The time taken for the heater to heat a certain amount of water from room temperature to 100 °C is as follows.

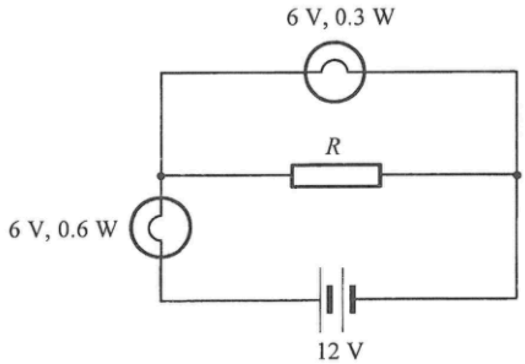
only X is connected	180 s
only Y is connected	300 s

How long does it take to heat the same amount of water from room temperature to 100 °C if both X and Y are connected ?

A. 105 s
B. 113 s
C. 154 s
D. 240 s

2026 Q26

26.



If both light bulbs work at their rated values as shown, find the resistance of R.

A. 30 Ω
B. 60 Ω
C. 90 Ω
D. 120 Ω

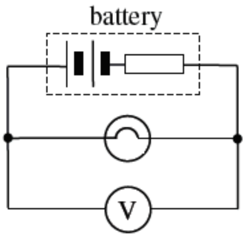
pp Q26

26. Two metal rods, X and Y, of uniform cross-sectional area are made of the same material and have the same volume. The length and resistance of X are l and R respectively. What is the resistance of Y if it has a length of $2l$?

A. $R/4$
B. $R/2$
C. $2R$
D. $4R$

pp Q27

27. The figure below shows a battery of e.m.f. 3.0 V and internal resistance 2.0 Ω is connected to a light bulb of resistance 10.0 Ω. A voltmeter of internal resistance 10 kΩ is connected in parallel with the light bulb. What is the reading of the voltmeter ?



A. 2.4 V
B. 2.5 V
C. 2.9 V
D. 3.0 V

pp Q28

28. In Figure (a), two identical resistors are connected in series to a cell of e.m.f. V and negligible internal resistance. The power dissipated by each resistor is P. If the two resistors are now connected in parallel as shown in Figure (b), what is the power dissipated by each resistor ?

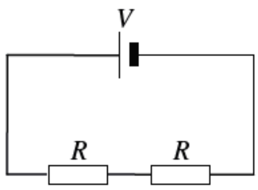


Figure (a)

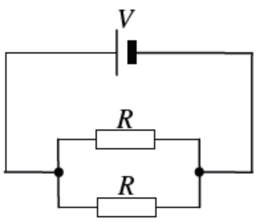
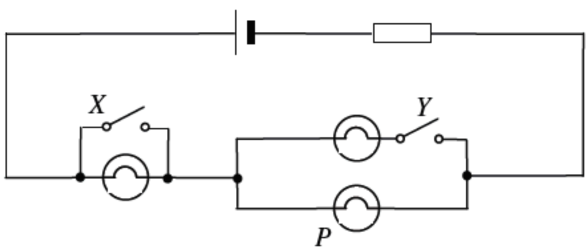


Figure (b)

A. $2P$
B. $4P$
C. $8P$
D. $16P$

pp Q29

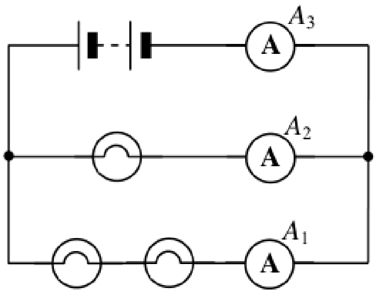
29. In the circuit below, three identical light bulbs are connected to a cell. Under what conditions will light bulb P have the maximum brightness ?



	Switch X	**Switch Y**
A.	closed	open
B.	closed	closed
C.	open	open
D.	open	closed

sap Q27

27.

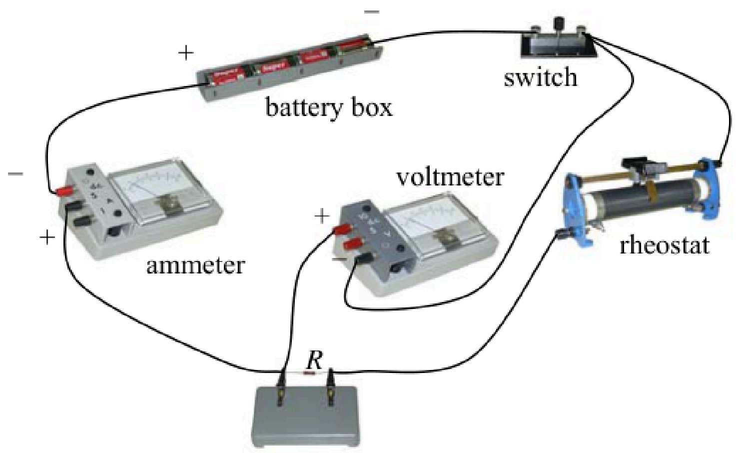


In the above circuit, the bulbs are identical. The reading of ammeter A_1 is 1 A. Find the readings of ammeters A_2 and A_3.

	Reading of A_2	**Reading of A_3**
A.	2 A	2 A
B.	2 A	3 A

sap Q30

30.



A student wants to measure the resistance of a resistor R and sets up the circuit shown. The student made which of these mistakes setting up the circuit ?

(1) The polarity of the ammeter was reversed.
(2) The polarity of the voltmeter was reversed.
(3) The voltmeter was connected across both R and the rheostat.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only